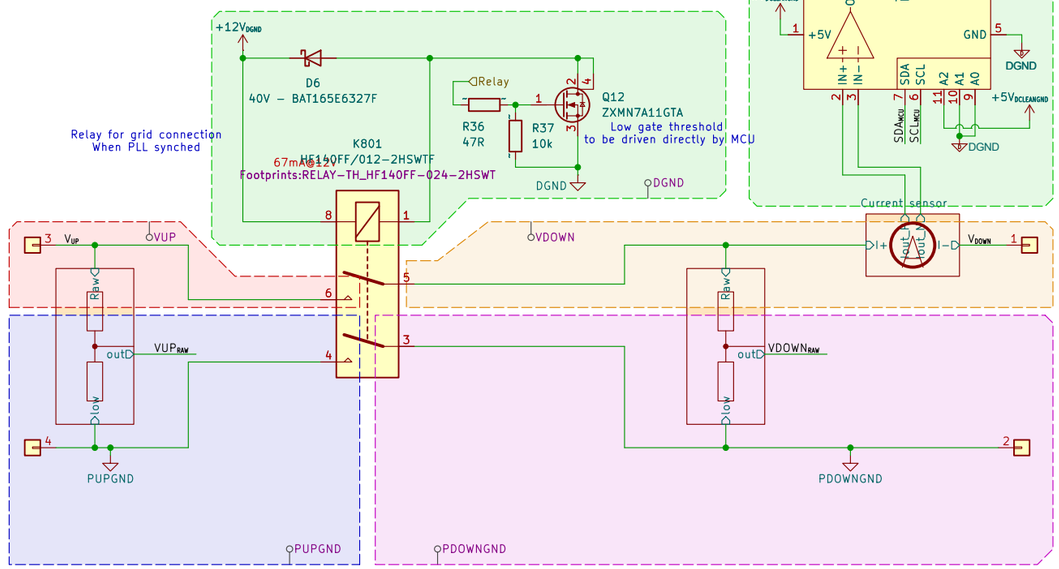
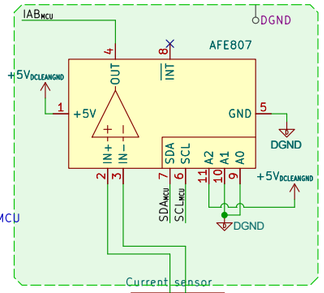
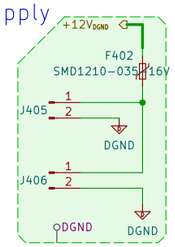


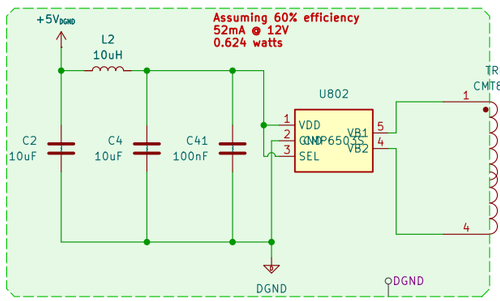


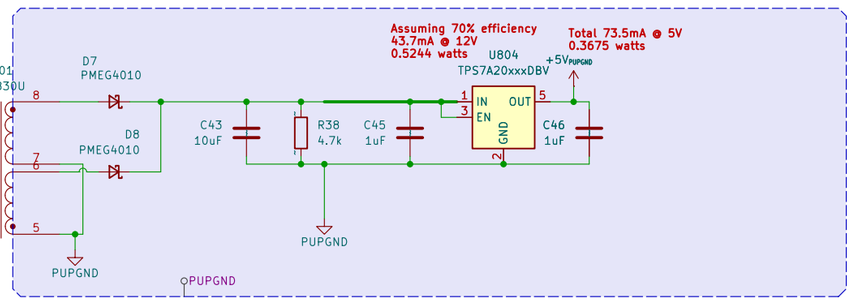


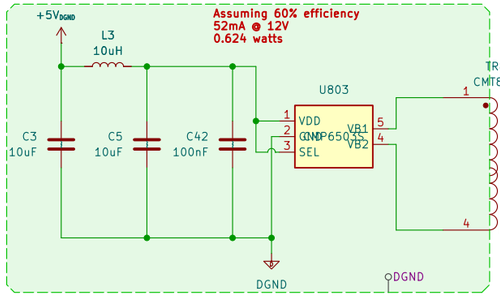
External power supply

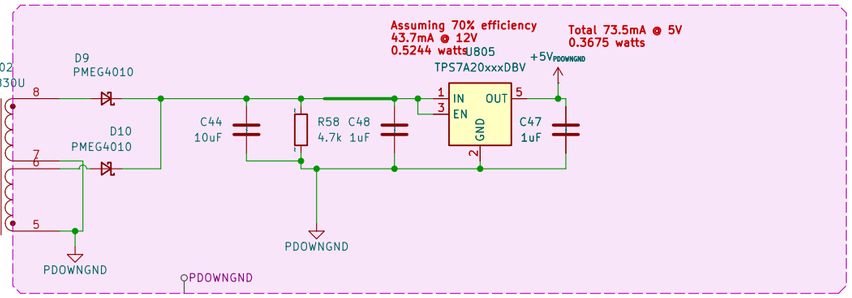


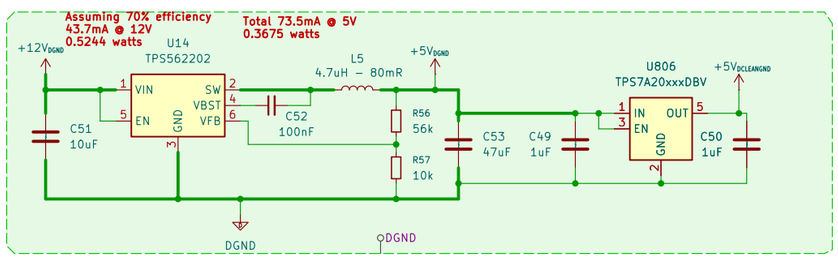
External power supply
- 12V
- Can be daisy chained
- Max current delivered to the daisy chain is 250mA











Total 31.3mA @ 12V
0.3756 watts

Powers low side AC inverter leg 1 and leg 2 2 x 0.6mA @ 12V

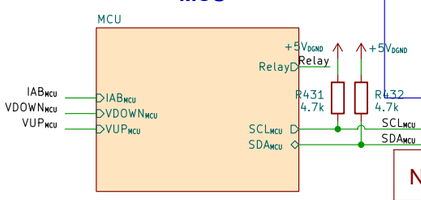
Powers high side AC inverter bootstrap leg 1 0.6mA @ 12V

Powers high side AC inverter bootstrap leg 2 0.6mA @ 12V

Powers digital isolators through TL431 2 x 7mA @ 12V (SN1 and SN2)

14.9mA @ 12V (GND_{PW})

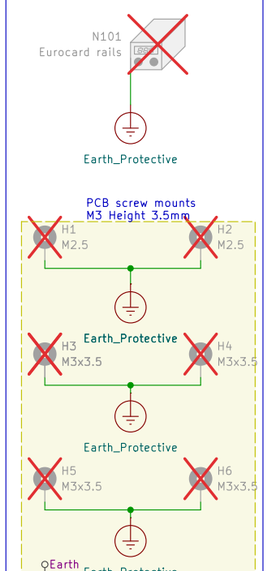
MCU



Backplane Signal ports



Mechanical mounts



NOTES :

All ceramic capacitor are X7R 50V unless specified
All resistor to are thick film 1/8Watt unless specified

Source repository : https://github.com/analuhaas/ext-board

Preliminary PREL, Good For Execution, GFE, As Built ASB
Major revisions are to be described thoroughly when pushed with git push command

Date	Rev.	Creator	Reviewer	Approver	Status
	B				
	A				
2026-08-11	-	ALH			

Title: External measurement board

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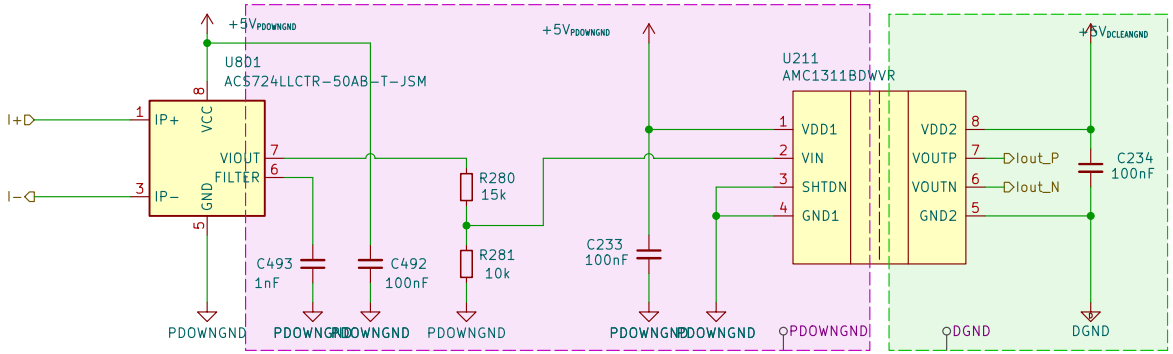
Licensed under
CERN-OHL-S V2.0

File: ext_board.kicad_sch

Size: A3 | Scale: | Sheet: /

KiCad E.D.A. 10.0.5 | Id: 1/7

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NOTES :

Source repository : https://github.com/analuhaas/ext-board

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Major revisions are to be described thoroughly when pushed with git push command

Date	Rev.	Creator	Reviewer	Approver	Status
	B				
	A				
	-				

Title:

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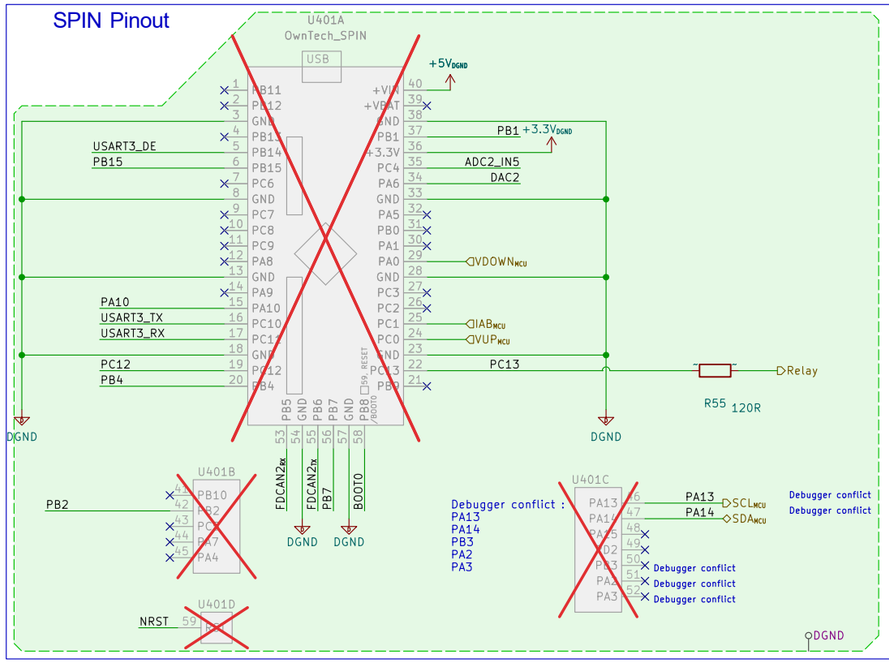
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File: current_sensor.kicad_sch

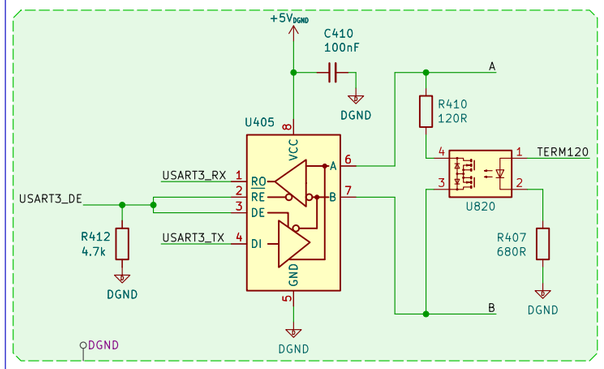
Size: A4 | Scale: | Sheet: /Current sensor/

KiCad E.D.A. 10.0.5 | Id: 2/7

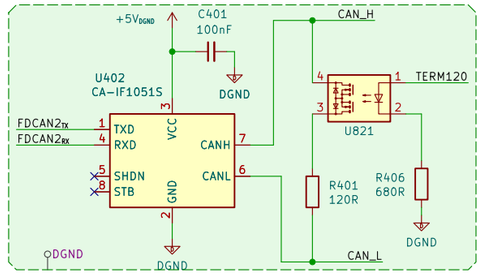
SPIN Pinout



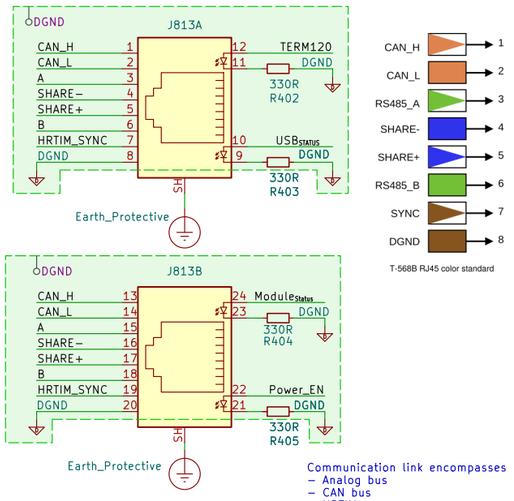
RS485 Transceiver



CAN Transceiver

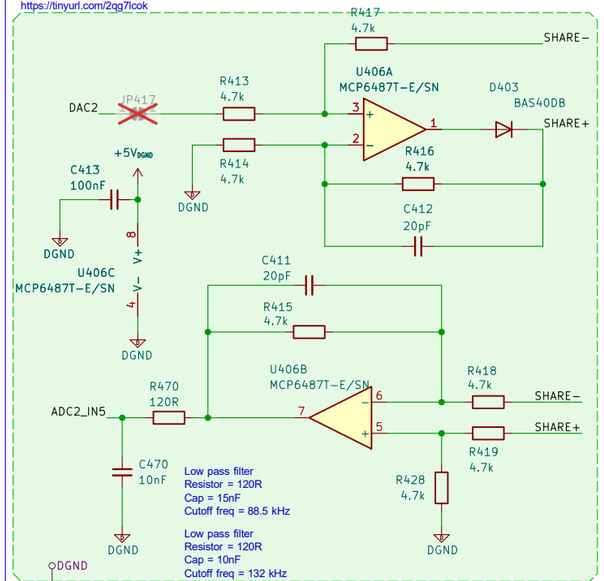


Communication RJ45 terminals



Analog bus

Current sharing analog bus. Highest voltage is imposing its voltage to the bus
Simulation available
https://tinyurl.com/2qg7lcok



HW configuration



NOTES :

Source repository : https://github.com/analuhaas/ext-board

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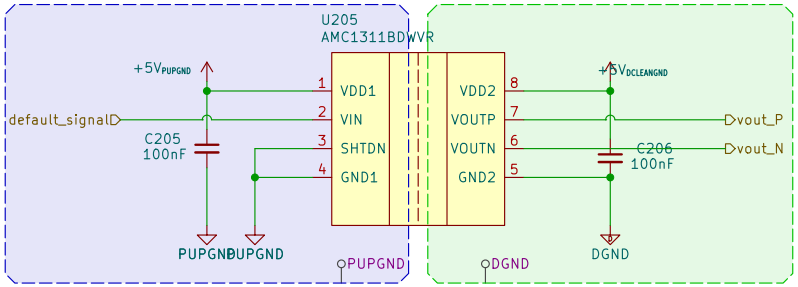
Date	Rev.	Creator	Reviewer	Approver	Status
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	A				
2025-10-13	-	JAL	LFL	LFL	GFE

Title: OWNVERTER V1.1

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File: Microcontroller.kicad_sch		
Size: A3	Scale:	Sheet: /MCU/
KiCad E.D.A. 10.0.5		Id: 3/7



NOTES :

Source repository : https://github.com/analuhaas/ext-board

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Major revisions are to be described thoroughly when pushed with git push command

Date	Rev.	Creator	Reviewer	Approver	Status
	B				
	A				
	-				

Title:

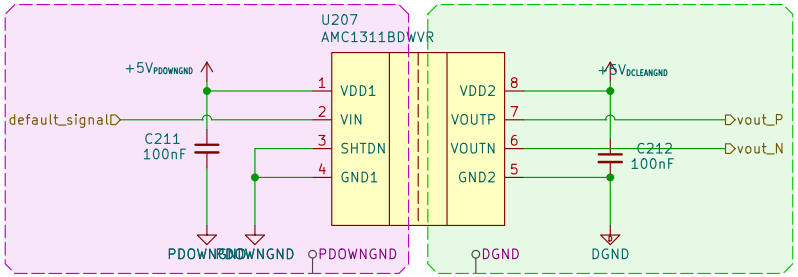
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File: Isolated_amplifier_UP.kicad_sch

Size: A4 | Scale: | Sheet: /Isolated amplifi

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NOTES :

Source repository : https://github.com/analuhaas/ext-board

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Major revisions are to be described thoroughly when pushed with git push command

Date	Rev.	Creator	Reviewer	Approver	Status
	B				
	A				
	-				

Title:

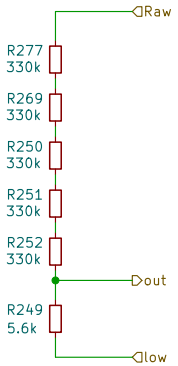
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File: Isolated_amplifier_DOWN.kicad_sch

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KiCad E.D.A. 10.0.5 | Id: 14/7



NOTES :

Source repository : https://github.com/analuhaas/ext-board

Preliminary PREL, Good For Execution, GFE, As Built ASB
Major revisions are to be described thoroughly when pushed with git push command

Date	Rev.	Creator	Reviewer	Approver	Status
	B				
	A				
	-				

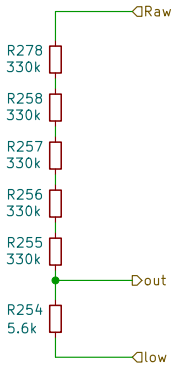
Title:

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File: Resistive_bridge.kicad_sch

Size: A4	Scale:	Sheet: /Resistive_bridge
KiCad E.D.A. 10.0.5		Id: 21/7



NOTES :

Source repository : https://github.com/analuhaas/ext-board

Preliminary PREL, Good For Execution, GFE, As Built ASB
Major revisions are to be described thoroughly when pushed with git push command

Date	Rev.	Creator	Reviewer	Approver	Status
	B				
	A				
	-				

Title:

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File: Resistive_bridge.kicad_sch

Size: A4	Scale:	Sheet: /Resistive_bridge
KiCad E.D.A. 10.0.5		Id: 22/7